A · PATIENT DEMOGRAPHICS & WRIST / HAND PRESENTATION PROFILE — INTAKE

Patient Name _____ Age _____ Sex ☐ M ☐ F Date _____

Occupation / Hand-Intensive Work / Sport _____ Dominant Hand ☐ R ☐ L Affected Side ☐ R ☐ L ☐ Bilat Onset ☐ Traumatic / FOOSH ☐ Overuse / Insidious

NPRS Now (circle) 0 1 2 3 4 5 6 7 8 9 10 Region ☐ Radial ☐ Ulnar ☐ Dorsal ☐ Palmar ☐ Digits Worst _____

Baseline Vitals (write-in): BP _____ HR _____ SpO_2 _____ Temp _____

B · SUSPECTED SCAPHOID FRACTURE, ACUTE CARPAL TUNNEL SYNDROME, CRPS & RED-FLAG TRIAGE — TRIAGE GATE

SUSPECTED SCAPHOID FRACTURE — ANATOMICAL SNUFFBOX, RULE & IMAGING — Mallee 2020

- ☐ **Epidemiology & exam (SR/MA, 75 studies)** (Carpenter 2014 Acad Emerg Med) [M]: **scaphoid = 70% of carpal fractures; occult fracture prevalence 12% to 57%, point estimate 25% pretest probability with non-diagnostic X-ray; only the absence of snuffbox tenderness helps (LR− 0.15)**
- ☐ **Anatomical snuffbox & single signs** (Mallee 2014 J Hand Surg Am · Parvizi 1998 J Hand Surg Br) [M]: **snuffbox tenderness the most sensitive test; low specificity → many overtreated; combining tests improved post-test probability · pain on thumb movement Sn 69%, Sp 66% (L)**
- ☐ **Clinical decision rule (n = 893, 7.6% fractures)** (Mallee 2020 Arch Orthop Trauma Surg) [M]: **sex, snuffbox swelling, snuffbox tenderness, painful ulnar deviation, painful axial thumb compression; internal validation Sn 97%, Sp 20%; 15% fewer unnecessary immobilisations and imaging, 50% lower risk of missing a fracture (not externally validated)**
- ☐ **Advanced imaging** (Bäcker 2020 J Wrist Surg · Mallee 2015 Cochrane) [M]: **X-ray + exam may miss up to 16%; MRI Sn 94.2%, Sp 97.7%; CT 81.5%, 96.0%; ultrasound 81.5%, 77.4% · Cochrane: CT Sn 0.72 (0.36–0.92), Sp 0.99 (0.71–1.00); MRI 0.88 (0.64–0.97), 1.00 (0.38–1.00); bone scan 0.99 (0.69–1.00), 0.86 (0.73–0.94)**

ACUTE CTS, CRPS & SAME-DAY RED FLAGS — Harden 2010

- ☐ **Acute carpal tunnel syndrome after DRF (n = 1,189)** (Leow 2021 Eur J Orthop Surg Traumatol) [L]: **51 (4.3%) developed acute CTS; median onset one week (range 1–12); pain out of proportion 13.7%, reduced power 9.8%; 10 of 51 (20%) resolved without intervention; 5 (10%) developed CRPS; type C fractures at higher risk**
- ☐ **CRPS — Budapest criteria (113 CRPS vs 47 neuropathic pain)** (Harden 2010 Pain) [M]: **IASP criteria Sn 1.00, Sp 0.41; Budapest clinical criteria Sn 0.99, Sp 0.68; Budapest research criteria Sp 0.79**
- ☐ **Pyogenic flexor tenosynovitis — Kanavel signs** [E]: **fusiform swelling of the digit, flexed resting posture, tenderness along the flexor sheath, pain on passive extension → same-day surgical referral**
- ☐ **Poor-outcome factors** (Pang 2007 J Bone Joint Surg Am) [L]: **age > 43 years; diabetes, peripheral vascular disease or renal failure; subcutaneous purulence; digital ischaemia; polymicrobial infection**
- ☐ **Other emergencies** [E]: **compartment syndrome; septic arthritis; high-pressure injection injury; fight bite (clenched-fist knuckle wound); acute CTS after fracture → same-day surgical referral**

☐ **Medication & Modality Safety:** Houghton 2010 (Guideline): electrophysical contraindications (malignancy, active DVT, pregnancy, electronic implants) • Kearns 2019 [E]: anticoagulants / coagulopathy — caution — possible increased incidence or severity of haematoma and bruising • Wise 2012 [M]: fluoroquinolones: tendonitis OR 4.3, rupture OR 2.0; stronger with oral glucocorticoids (OR 9.1) and age > 60 (OR 8.3)

TRIAGE DISPOSITION: ☐ **Cleared — proceed** ☐ Imaging required (suspected scaphoid / fracture) ☐ Urgent referral (acute CTS after fracture / suspected CRPS)
☐ **Same-day surgical referral (Kanavel / compartment / septic / injection / fight bite)**

C · CLINICAL PATHWAY ROUTER & WRIST / HAND SUBGROUP NAVIGATION — RAPID

CLINICAL PATHWAY ROUTER — Tick route → jump to matching pages

☐ A · NERVE (CTS): 1 → 2 → 4 → 6 → 7 ☐ B · TENDON / OA / INSTABILITY: 1 → 2 → 3 → 4 → 8 ☐ C · FRACTURE / POST-OP: 1 → 3 → 5 → 8 → 9

D · SUBJECTIVE HISTORY, SYMPTOM BEHAVIOUR & WRIST / HAND BODY CHART — SUBJECTIVE

SYMPTOM LOCATION & CLINICAL PATTERN — Clinical Signpost

- ☐ **Night paraesthesia thumb–index–middle, shaking relieves:** Flick sign ▸ / Wainner ▸ → **Carpal tunnel syndrome (p2/p4/p6)**
- ☐ **Radial styloid pain with thumb use:** WHAT test ▸ → **De Quervain (p2/p4/p6)**
- ☐ **Thumb-base pain on pinch / jar opening:** Grind test ▸ → **Thumb CMC OA (p2/p4/p6)**
- ☐ **Ulnar wrist pain on rotation / grip:** Fovea sign ▸ / DRUJ test ▸ → **TFCC / DRUJ (p2/p4)**

AGGRAVATING, EASING & IRRITABILITY PROFILE — Behaviour

Aggravating _____ Easing _____

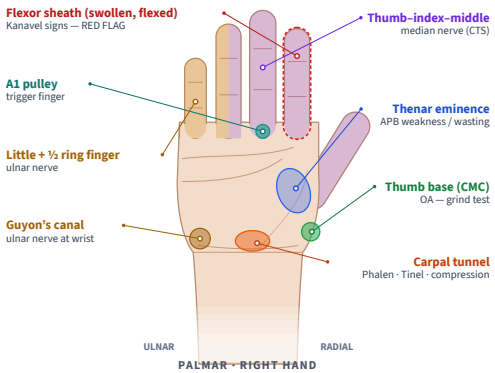
Irritability: ☐ Low ☐ Moderate ☐ High (rest/night pain)

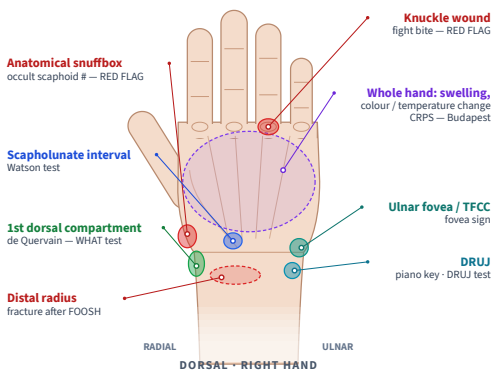
MECHANISM & OCCUPATIONAL / SPORT RISK — History

Mechanism: ☐ FOOSH / fall ☐ Repetitive gripping / keyboard ☐ Crush / laceration ☐ Pregnancy / postpartum

Factors: ☐ Diabetes ☐ Prior cast / fixation ☐ Locking / triggering ☐ Colour / temperature change

BODY CHART — WHERE IS THE PAIN? — Clinical signposts [E]





Mark on the chart: shade = pain · //// = pins & needles · ×× = numbness · → = radiation · write 0–10 beside each area

Anatomical snuffbox tenderness after FOOSH → suspected scaphoid pathway (imaging). **Swollen, flexed digit** → Kanavel signs (same day).

SUBJECTIVE HISTORY · MECHANISM · PATIENT GOALS (PSFS)

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

A · WRIST & HAND DIFFERENTIAL DIAGNOSIS MATRIX (NEURAL, RADIAL, ULNAR, CARPAL & DIGITAL)

DIFFERENTIAL

CONDITION	HALLMARK HISTORY & PAIN LOCATION	KEY PHYSICAL FINDINGS & PROVOCATION	EVIDENCE & ATLAS
Carpal tunnel syndrome	Night paraesthesia in thumb, index, middle; shaking the hand relieves symptoms	Wainner cluster ▸, Katz diagram / median hypalgesia, Phalen ▸ / Tinel ▸ / compression ▸	H Confirmed 2.7% (2.1–3.4) (Atroshi 1999)
De Quervain tenosynovitis ▸	Radial styloid pain with thumb use, lifting and gripping	WHAT test ▸ (Sn 0.99, Sp 0.29), Finkelstein ▸, Eichhoff ▸	L Goubau 2014
Thumb CMC OA ▸	Thumb-base pain on pinch, key turning and jar opening	Grind test ▸ (LR+ 2.65 and 6.00); a negative test does not exclude	L Merritt 2010
Trigger finger	Catching or locking of a digit, worse in the morning	Tender nodule at the A1 pulley; triggering on active composite flexion	E Clinical differentiation
TFCC / ulnar-sided pain ▸	Ulnar wrist pain with rotation, grip and ulnar deviation	Fovea sign ▸ (Sn 95.2%, Sp 86.5%), TFCC compression ▸, ulnar grinding ▸	M Tay 2007
DRUJ instability ▸	Painful clunk or weakness with forearm rotation after a fall or DRF	DRUJ test ▸: dorso-palmar laxity vs the other side	E Clinical differentiation
Scapholunate instability ▸	Dorsal central wrist pain after FOOSH; clunk with loaded movement	Watson scaphoid shift test ▸; midcarpal test ▸	E Clinical differentiation
Scaphoid fracture / non-union	FOOSH with radial wrist pain; persisting pain after a normal X-ray	Snuffbox tenderness (absence LR− 0.15); Mallee 2020 rule; MRI	M Carpenter 2014
Distal radius fracture	FOOSH with deformity, swelling and pain over the distal radius	X-ray; monitor median nerve (acute CTS 4.3%)	L Leow 2021
CRPS	Pain out of proportion; colour, temperature, sweating or swelling change	Budapest clinical criteria (Sn 0.99, Sp 0.68)	M Harden 2010
Hand OA	Aching and stiffness of DIP / PIP / thumb-base joints	Nodal swelling, reduced grip; track with AUSCAN	E Clinical differentiation

B · CARPAL TUNNEL PREVALENCE & NATURAL HISTORY + RADIAL / ULNAR TEST ACCURACY

PROGNOSIS

CTS PREVALENCE & NATURAL HISTORY

Atroshi 1999

- ☐ **General population (n = 2,466)** (Atroshi 1999 JAMA) H: **median-territory symptoms 14.4% (13.0–15.8); clinically certain CTS 3.8% (3.1–4.6); electrophysiological median neuropathy 4.9% (4.1–5.8); clinically + electrophysiologically confirmed 2.7% (2.1–3.4); 1 in 5 symptomatic people have CTS; blue-collar 3.5% vs white-collar 1.7%; median neuropathy in 18.4% of asymptomatic controls**
- ☐ **Untreated CTS (n = 80)** (Padua 1998 Ital J Neurol Sci) L: **worsening inversely related to severity; about one-third of mild and moderate cases improved; extreme cases never improved**

DE QUERVAIN, CMC & TFCC ACCURACY

Radial / Ulnar

- ☐ **WHAT vs Eichhoff (n = 100, ultrasound reference)** ▸ (Goubau 2014 J Hand Surg Eur) L: **WHAT test Sn 0.99, Sp 0.29, PPV 0.95, NPV 0.67 — more accurate than Eichhoff's test (often wrongly called Finkelstein's)**
- ☐ **Thumb CMC grind test** ▸ (Merritt 2010 J Hand Ther) L: **κ 0.48; Sn 42% and 53%, Sp 80% and 93% (two testers); LR+ 2.65 and 6.00; LR− 0.59 and 0.62 — confirms but a negative test does not exclude**
- ☐ **Ulnar fovea sign** ▸ (Tay 2007 J Hand Surg Am · Chang 2019 J Hand Surg Asian Pac) M: **Sn 95.2%, Sp 86.5% for foveal disruption and / or ulnotriquetral ligament injury · outcomes (L): non-surgical care 54% improved pain; surgery 83%**

C · PERSISTENT WRIST PAIN PROGNOSIS & POST-DRF PSYCHOLOGICAL SCREEN

PROGNOSIS

PERSISTENT WRIST PAIN (PROSSER 2012)

Arthroscopy Cohort

- ☐ **Persistent wrist pain** (Prosser 2012 J Hand Ther) L: **improved on average by approximately 50% at one year, but most had some pain and disability; duration of pain and baseline PRWHE explained 24% of 1-year PRWHE**
- ☐ **DRF — psychological screen** (Mahdavi 2022 Musculoskelet Sci Pract) L: **PRWE pain or function ≥ 18/50 at 6 months screens for pain catastrophizing, distress and fear (AUC ≥ 0.88)**

Primary Working Diagnosis: _____
Secondary / Contributing Factors: _____
Yellow Flags / Fear / Catastrophizing: _____

DIFFERENTIAL REASONING · PROGNOSTIC COUNSELLING · RISK-FACTOR NOTES

Clinician _____ **Date** _____

Underline = Atlas card · ▸ = video · Evidence: H high M moderate L low E expert · Confidential · v2026.10

A · WRIST, FOREARM, THUMB & FINGER RANGE OF MOTION WRITE-IN TABLE

GONIOMETRY

MOTION (AROM ▸ / PROM ▸)	AFFECTED AROM	AFFECTED PROM	UNAFFECTED	END-FEEL / PAIN
Wrist Flexion				☐ Firm ☐ Caps ☐ Pain
Wrist Extension				☐ Firm ☐ Bony ☐ Pain
Radial Deviation				☐ Firm ☐ Bony ☐ Pain
Ulnar Deviation				☐ Firm ☐ Caps ☐ Pain
Forearm Pronation				☐ Firm ☐ Pain
Forearm Supination				☐ Firm ☐ Pain
Thumb CMC / MCP / IP & Opposition				☐ Full ☐ Caps ☐ Pain
Fingers MCP / PIP / DIP & Composite Fist				☐ Full ☐ Caps ☐ Pain

☐ **Wrist goniometry (multicentre)** (LaStayo 1994 Phys Ther) [M]: **flexion ICC within tester .86 / .87 / .92 (radial / ulnar / dorsal alignment), extension .80 / .80 / .84; between testers flexion .88 / .89 / .93; volar/dorsal alignment gives least error** • Soucie 2011 [M]: significant age- and sex-related variation — compare with the other side and the patient's own baseline

☐ **Resisted isometric testing** ▸ [E]: wrist flexion / extension, radial / ulnar deviation, thumb and finger flexors / extensors — pain on contraction vs weakness

B · GRIP & PINCH DYNAMOMETRY & ISOMETRIC STRENGTH

DYNAMOMETRY · TORQUE

GRIP NORMS & DEVICE RULES

Grip Ratio Calc

☐ **Grip norms:** Wang 2018 [M]: mean grip 49.7 kg (men 25–29 y) to 18.7 kg (women 75–79 y) • Mutalib 2022 [L]: GripAble reads 69% of Jamar (ICC 0.906) — **devices must not be interchanged**

☐ **HHD rule (E):** Same device, same position, same strap / lever-arm point and tester; compare with other side.

MEASURE	LEVER (CM)	AFFECTED (N/KG)	AFF (N·M)	UNAFF (N·M)	LSI (%)
Wrist Extension (HHD)					
Wrist Flexion (HHD)					
Grip (kg)	—		—	—	
Key pinch (kg)	—		—	—	

C · SENSORY & NEURO SCREEN, VASCULAR, NEURODYNAMIC & OEDEMA WRITE-IN

NEURO & OEDEMA

NERVE [E]	SENSORY TERRITORY	MOTOR KEY TEST	FINDING (L / R)
Median	Palmar thumb, index, middle & radial half of ring finger	APB thumb abduction; opposition	
Ulnar	Little finger & ulnar half of ring finger; hypothenar	Interossei; Froment; Wartenberg	
Radial	Dorsal first web space & dorsoradial hand	Wrist & finger extension	

SENSORY DETAIL & NEURODYNAMICS [E]

2-point discrimination: _____ **Monofilament:** _____
ULNT1 (median) ▸ **L / R:** _____ **Hypalgesia map:** _____

VASCULAR & OEDEMA [E]

Allen test ▸: ☐ Radial patent ☐ Ulnar patent ☐ Delayed refill
Figure-of-eight A / U: _____ **Volumetry A / U:** _____

OBJECTIVE EXAMINATION · GONIOMETRY · GRIP / PINCH · SENSORY & OEDEMA NOTES

A · CARPAL TUNNEL SYNDROME TESTS, RATIONAL CLINICAL EXAM & CLINICAL PREDICTION RULE

CTS TESTS

TEST	TECHNIQUE & POSITIVE CRITERION	VERIFIED DIAGNOSTIC PERFORMANCE	RESULT
Phalen's test ▸ & reverse Phalen	Sustained full wrist flexion (or extension) → median-territory paraesthesia	Phalen **Sn 68%, Sp 73%** (MacDermid 2004) M · reverse E	☐ – ☐ +
Tinel sign (wrist) ▸	Tap over median nerve at the carpal tunnel → distal tingling	**Sn 50%, Sp 77%** (MacDermid 2004) M	☐ – ☐ +
Carpal compression ▸	Thumb pressure over the carpal tunnel → median symptoms	**Sn 64%, Sp 83%** (MacDermid 2004) M	☐ – ☐ +
Flick sign ▸ & hand elevation ▸	Shaking relieves symptoms; arms overhead reproduce symptoms	Wainner / CPG item L · elevation E	☐ – ☐ +
Katz diagram, hypalgesia & thumb abduction	Hand symptom diagram; pinprick median vs little finger; resisted APB	For: hypalgesia **LR 3.1 (2.0–5.1)**, classic / probable Katz **2.4 (1.6–3.5)**, weak thumb abduction **1.8 (1.4–2.3)**; against: unlikely Katz **0.2 (0.0–0.7)**, normal thumb abduction **0.5 (0.4–0.7)** (D'Arcy 2000) M	☐ – ☐ +
Wainner CPR ▸ Calc	Shaking relieves; wrist-ratio index > .67; SSS > 1.9; reduced median sensory field of digit 1; age > 45	**LR 18.3** when all 5 positive; post-test probability changes up to **56%** (not validated) (Wainner 2005) L	☐ – ☐ +
CTS-6	Delphi-derived weighted clinical criteria	Correlation with clinician panel **0.71** (Graham 2006) L	☐ – ☐ +

☐ **JOSPT CTS CPG — diagnosis** (Erickson 2019 JOSPT CTS CPG) (Guideline): **"Clinicians should assess and document patient age (older than 45 years), whether shaking their hands relieves their symptoms, sensory loss in the thumb, the wrist ratio index (greater than 0.67), and scores from the Boston Carpal Tunnel Questionnaire symptom severity scale (CTQ-SSS) (greater than 1.9). The presence of more than 3 of these clinical findings has shown acceptable diagnostic accuracy."**

☐ **Single findings** (D'Arcy 2000 JAMA) M: **Phalen, Tinel, nocturnal paraesthesia, thenar atrophy, 2-point / vibration / monofilament have little or no value**; 2-point discrimination and APB weakness / atrophy specific but not sensitive (MacDermid 2004)

B · DE QUERVAIN, THUMB CMC, TFCC / DRUJ & CARPAL INSTABILITY TESTS

SPECIAL TESTS

TEST	TECHNIQUE & POSITIVE CRITERION	VERIFIED DIAGNOSTIC PERFORMANCE	RESULT
WHAT test ▸, Finkelstein ▸ & Eichhoff ▸	Wrist hyperflexion + resisted thumb abduction / extension; passive thumb traction; thumb-in-fist ulnar deviation	WHAT **Sn 0.99, Sp 0.29, PPV 0.95, NPV 0.67** (Goubau 2014) L	☐ – ☐ +
CMC grind test ▸	Axial load + rotation of first metacarpal on trapezium	**κ 0.48; LR+ 2.65 and 6.00; LR− 0.59 and 0.62** (Merritt 2010) L	☐ – ☐ +
Ulnar fovea sign ▸	Press into the fovea between ulnar styloid and FCU	**Sn 95.2%, Sp 86.5%** (Tay 2007) M	☐ – ☐ +
TFCC compression ▸ & ulnar grinding ▸	Ulnar deviation + axial load ± rotation of the carpus	Ulnocarpal provocation E	☐ – ☐ +
DRUJ test ▸	Dorso-palmar translation of the ulnar head vs stabilised radius	Laxity vs other side E	☐ – ☐ +
Watson scaphoid shift ▸	Thumb on scaphoid tubercle; ulnar → radial deviation; release → clunk	Scapholunate instability E	☐ – ☐ +
LT ballottement ▸ & midcarpal test ▸	Shuck triquetrum on lunate; palmar load + ulnar deviation → catch-up clunk	Carpal instability E	☐ – ☐ +

C · VALIDATED WRIST & HAND PROM DASHBOARD (PRWE, QUICKDASH / DASH, BCTQ, NPRS, MHQ, AUSCAN, PSFS)

PROMS

OUTCOME SCALE	SCORING RULE & RANGE	VERIFIED RELIABILITY, MDC & MCID (POPULATION)	INITIAL	FOLLOW-UP
PRWE Calc	15 items (0–10): Pain (5 items, 0–50) + Function (10 items ÷ 2, 0–50) = 0–100	ICC **> 0.90** (MacDermid 1998) M · Surgical DRF: **71.8 → 34.3 (6 weeks) → 21.2 (12 weeks); MCID 43.1 (6 weeks), 56.0 (12 weeks)** (Huyke-Hernández 2023) L		
QuickDASH / DASH Calc	11 items (1–5, ≥ 10 answered): ((sum ÷ n) − 1) × 25	Upper limb: ICC **0.91 / 0.93**; MDC90 **12.85 / 10.81**; MCID **15.91** (QuickDASH), **10.83** (DASH) (Franchignoni 2014) M		
BCTQ Calc · **NPRS**	SSS mean of 11 items; FSS mean of 8 items (1–5) · NPRS 0–10	Women with CTS: MCID BCTQ **0.23** (functional status), **0.64** (symptom severity); NPRS **1.5** (mean), **2.5** (worst pain) (López-de-Uralde-Villanueva 2024) L		
MHQ	Domain scores (function, ADL, work, pain, aesthetics, satisfaction)	Older adults after DRF: five domains fit Rasch model; interpret domain scores rather than the summary score (Jayaram 2021) L		
AUSCAN · **PSFS** Calc	AUSCAN pain / stiffness / function · PSFS mean 0–10	Clinical tracking scales E		

SPECIAL TESTS SYNTHESIS · BASELINE PROM SCORES · FUNCTIONAL GOAL SETTING

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: H high M moderate L low E expert · Confidential · v2026.10

A · DISTAL RADIUS FRACTURE: FIXATION TRIALS (WRIST, DRAFFT) & REHABILITATION

POST-FRACTURE

POST-OP RESTRICTIONS ARE SURGEON-DIRECTED [E]

FIXATION TRIALS — DRAFFT

- ☐ **DRF ≥ 60 years — WRIST trial** (Chung 2020 Plast Reconstr Surg) [H]: **12-month MHQ: volar plate vs external fixation 3 points (97.5% CL −6.0 to 11.5), vs pinning −0.14 (−9.2 to 8.9); at 6 weeks plate better than external fixation by 19, pinning by 11, casting by 7 (p = 0.03); malunion in 48% of casting participants**
- ☐ **K-wires vs volar plate (DRAFFT, n = 461)** (Costa 2014 BMJ) [H]: **no clinically relevant PRWE difference at 3, 6 or 12 months (−1.3, −4.5 to 1.8; P = 0.40); DASH at 12 months −3.2 (−6.5 to 0.0)**
- ☐ **DRAFFT 5-year** (Costa 2019 Bone Joint J) [H]: **PRWE 8.3 (wires) vs 11.3 (plate) (−6.99 to 0.99; p = 0.139) — no clinically relevant difference**

REHABILITATION AFTER DRF — Handoll 2015

- ☐ **Rehabilitation (Cochrane, 26 trials)** (Handoll 2015 Cochrane) [M]: **"not enough evidence available to determine the best form of rehabilitation for people with wrist fractures"**
- ☐ **Advice + exercise vs no intervention** (Kay 2008 Aust J Physiother) [M]: **advice and exercise program gave "some additional benefits over no intervention"**
- ☐ **Volar plate — early vs delayed motion (RCT, n = 48)** (Laohaprasitiporn 2022 Sci Rep) [M]: **no significant differences in function, radiographs or complications**
- ☐ **Manual therapy (SR/MA)** (Gutiérrez-Espinoza 2022 J Man Manip Ther) [L]: **exercise + MWM vs exercise at 12 weeks: PRWE −10.2 (p = 0.02), DASH −9.86 (p = 0.0001); manual lymph drainage reduced oedema (e.g. −17.96 ml at 17–21 days)**

B · CARPAL TUNNEL RELEASE, SCAPHOID, FLEXOR TENDON, THUMB CMC, TRIGGER FINGER & CRPS PREVENTION

POST-OP

CARPAL TUNNEL RELEASE & SCAPHOID WAIST — Peters 2016

- ☐ **Surgery vs night splint (RCT, n = 176)** (Gerritsen 2002 JAMA) [H]: **success at 3 months 80% vs 54% (difference 26%, 12–40%); at 18 months 90% vs 75% (15%, 3–27%)**
- ☐ **Surgery vs non-surgical therapy (no denervation)** (Jarvik 2009 Lancet) [H]: **"surgery is useful for patients with carpal tunnel syndrome without denervation"**
- ☐ **Rehab after release (Cochrane, 22 trials, 1521 pts)** (Peters 2016 Cochrane) [M]: **limited, low-quality evidence; decision "should be based on the clinician's expertise, the patient's preferences and the context"**
- ☐ **Scaphoid waist — surgery vs cast (SWIFFT, n = 439)** (Dias 2020 Lancet) [H]: **no significant PRWE difference at 52 weeks (surgery adjusted mean 11·9, 9·2–14·5)**

TENDON, CMC, TRIGGER FINGER & CRPS — Hand Surgery

- ☐ **Flexor tendon repair (SR, 34 studies)** (Starr 2013 J Hand Surg Am) [M]: **early passive motion: rupture 4%, reduced ROM 9%; early active motion: rupture 5%, reduced ROM 6%**
- ☐ **Thumb CMC OA surgery (Cochrane, 11 RCTs)** (Wajon 2015 Cochrane) [L]: **LRTI vs trapeziectomy alone: pain −2.8 mm (−9.8 to 4.2); adverse events 19 vs 10 per 100 (RR 1.89, 0.96–3.73) — LRTI may add nothing**
- ☐ **Trigger finger — release vs injection (RCTs)** (Sato 2012 Rheumatology · Zyluk 2011 J Hand Surg Eur) [M]: **percutaneous and open release similar and superior to steroid injection for cure and relapse; percutaneous release lower recurrence than injection**
- ☐ **CRPS after wrist fracture — vitamin C (RCT, 416 pts)** (Zollinger 2007 J Bone Joint Surg Am) [M]: **CRPS 2.4% vitamin C vs 10.1% placebo (p = 0.002); 500 mg: 1.8% (RR 0.17, 0.04–0.77)**
- ☐ **Vitamin C (MA, 3 RCTs)** (Evaniew 2015 J Orthop Trauma) [M]: **"fails to demonstrate a significant benefit"**

C · WOUND / NEURO / CAST-COMPLAINT RE-CHECK BOX & OPERATIVE DETAILS LOG

OPERATIVE LOG

Re-Check [E]: ☐ Wound erythema / drainage / infection ☐ New median / ulnar / radial deficit ☐ Cast too tight / burning pain / swelling ☐ Hardware or tendon complaint

Early cast-related complaints (Zollinger 2007 J Bone Joint Surg Am) [M]: **predicted CRPS (RR 5.35, 2.13–13.42)** → check cast and screen with Budapest criteria

Procedure / Fixation / Repair Zone _____ **Surgery Date** _____ **Surgeon** _____

Surgeon-Directed Restrictions & Orthosis / Cast Orders _____

Median / Ulnar / Radial Status _____ **Target RTW / Function Milestone** _____

Criteria to progress (E): ☐ Surgeon clearance verified ☐ Wound healed & no infection ☐ Pain controlled (zone ≤ 5/10) ☐ Target PROM arc achieved ☐ Strength baseline recorded ☐ Patient confident

POST-OPERATIVE COMMUNICATION · SURGEON ORDERS · ORTHOSIS / CAST NOTES

A · CARPAL TUNNEL SYNDROME: CORTICOSTEROID INJECTION, NIGHT SPLINTING & ULTRASOUND

INJECTION & ORTHOSIS

CORTICOSTEROID INJECTION

Hofer 2021

- ☐ **Steroid injection (Cochrane)** (Marshall 2007 Cochrane) [M]: **no better than anti-inflammatory + splint at 8 weeks or laser at 6 months; two injections no better than one**
- ☐ **Methylprednisolone vs placebo (RCT, n = 111)** (Atroshi 2013 Ann Intern Med) [H]: **10 weeks: symptom score −0.64 (80 mg) and −0.88 (40 mg) vs placebo; no difference at 1 year; surgery within 1 year 73%, 81%, 92% (80 mg, 40 mg, placebo)**
- ☐ **Same trial — 5 years** (Hofer 2021 JAMA Netw Open) [H]: **symptom-score change vs placebo 0.14 (−0.17 to 0.45) for 80 mg and 0.12 (−0.19 to 0.43) for 40 mg**

SPLINTING & THERAPEUTIC ULTRASOUND

Page 2012

- ☐ **Splinting (Cochrane 2012)** (Page 2012 Cochrane) [L]: **night splint vs no treatment: overall improvement at 4 weeks RR 3.86 (2.29–6.51), low quality; neutral vs extension splint RR 2.43 (1.12–5.28), very low quality**
- ☐ **Splinting (Cochrane 2023)** (Karjalainen 2023 Cochrane) [L]: **12 weeks may not be better than six weeks; six months may be better than six weeks (low certainty)**
- ☐ **Therapeutic ultrasound (Cochrane)** (Page 2013 Cochrane) [L]: **one low-quality trial (68 pts): RR 2.36 (1.40 to 3.98) for improvement at 7 weeks; overall poor-quality evidence**

B · JOSPT CTS CPG ORTHOSIS, PHYSICAL-AGENT & ERGONOMICS QUOTES + DE QUERVAIN, CMC OA, TRIGGER FINGER & CRPS

CPG & EPAS

JOSPT CTS CPG (ERICKSON 2019)

JOSPT 2019

- ☐ **Orthoses & education** (Erickson 2019 JOSPT CTS CPG) (Guideline): **"Clinicians should recommend a neutral-positioned wrist orthosis worn at night for short-term symptom relief and functional improvement"; "may suggest adjusting wear time to include daytime, symptomatic, or full-time use when night-only use is ineffective"; "may add patient education on pathology, risk identification, symptom self-management, and postures/activities that aggravate symptoms"; "should recommend an orthosis for women experiencing CTS during pregnancy"**
- ☐ **Physical agents** (Erickson 2019 JOSPT CTS CPG) (Guideline): **"may recommend a trial of superficial heat"; "may recommend the application of microwave or shortwave diathermy"; "may offer a trial of interferential current"; "may perform phonophoresis"; "should not use low-level laser therapy or other types of nonlaser light therapy"; "should not use thermal ultrasound"; "should not use iontophoresis"; "should not use or recommend the use of magnets"**
- ☐ **Ergonomics** (Erickson 2019 JOSPT CTS CPG · O'Connor 2012 Cochrane) (Guideline): **"may educate their patients regarding the effects of mouse use on carpal tunnel pressure"; "may recommend keyboards with reduced strike force" · ergonomic equipment: insufficient evidence to determine benefit or harm (L)**

DE QUERVAIN, THUMB CMC OA, TRIGGER FINGER & CRPS

Orthoses & Injection

- ☐ **De Quervain — injection ± thumb spica (RCT)** (Ippolito 2020 Hand (N Y)) [L]: **outcomes comparable except radial wrist pain resolution better with injection alone (100% vs 64%)**
- ☐ **Thumb CMC OA — splints (SR/MA)** (Buhler 2019 Osteoarthritis Cartilage) [M]: **3–12 months: pain SMD 0.7 (1.04, 0.35), function SMD 0.42 (0.77, 0.08); low-quality evidence; no short-term effect**
- ☐ **Thumb base OA — combined care (RCT, n = 204)** (Deveza 2021 JAMA Intern Med) [H]: **splint + exercise + topical NSAID + education vs education: FIHOA −1.7 at 6 weeks, −2.4 (−3.5 to −1.3) at 12 weeks; pain −8.6 mm (−15.2 to −2.0)**
- ☐ **Trigger finger — steroid injection (Cochrane, 2 RCTs)** (Peters-Veluthamaningal 2009 Cochrane) [L]: **steroid + lidocaine superior to lidocaine alone ("silver" level evidence)**
- ☐ **CRPS — graded motor imagery / mirror therapy** (Moseley 2004 Pain · Méndez-Rebolledo 2017 J Back Musculoskelet Rehabil) [L]: **GMI RCT in chronic CRPS1 (13 pts) effective; SR of 6 trials (171 pts) GMI / mirror therapy improve pain and function in CRPS 1**

C · BLOOD FLOW RESTRICTION (BFR) AFTER DISTAL RADIUS FRACTURE

BFR · DRF

- ☐ **BFR in distal radius fracture:** Cancio 2019 [L] (closed management RCT): greater pain reduction over 8 weeks (p = 0.03); all patients tolerated BFR without early termination • Sgromolo 2020 [L] (after fixation RCT): greater reduction in pain with activity (p = 0.03) and PRWE (p = 0.01) over 8 weeks
- ☐ **Protocol (Patterson 2019 Guideline):** 40 to 80 % of arterial occlusion pressure (AOP), set from a measured AOP; 20–40 % 1RM; 75 repetitions as 30-15-15-15; rest 30–60 s; 2–3 sessions per week • **Cuff on proximal arm only** [E]

D · ORTHOSIS, INJECTION & ELECTROPHYSICAL AGENT DELIVERY LOG

PARAMETER LOG

DATE	ORTHOSIS / INJECTION / MODALITY	POSITION, WEAR TIME / DOSE / PARAMETERS	PRE / POST NPRS OR BCTQ	PATIENT RESPONSE

INJECTION COUNSELLING · ORTHOSIS FIT & WEAR SCHEDULE · MODALITY NOTES

A · CARPAL TUNNEL SYNDROME: EXERCISE, MOBILISATION, NEURAL MOBILISATION & MANUAL THERAPY VS SURGERY

MANUAL THERAPY

CTS MANUAL THERAPY EVIDENCE

Fernández-de-las-Peñas 2015

- ☐ **Exercise & mobilisation (Cochrane, 16 trials)** (Page 2012 Cochrane) [L]: **limited, low-quality evidence; decision should be based on clinician's expertise and patient preference**
- ☐ **Neural mobilisation (SR/MA, 40 RCTs)** (Basson 2017 JOSPT) [M]: **in CTS most clinical outcomes not improved (P > .11), some positive neurophysiological effects**
- ☐ **Manual therapy vs surgery (RCT, 120 women)** (Fernández-de-las-Peñas 2015 J Pain) [H]: **PT better at 1 and 3 months: mean pain −2.0 (−2.8 to −1.2) and −1.3 (−2.1 to −.6); worst pain −2.9 and −2.0; similar at 6 and 12 months**

JOSPT CTS CPG — MANUAL THERAPY (ERICKSON 2019)

JOSPT 2019

- ☐ **Cervical spine & upper extremity** (Erickson 2019 JOSPT CTS CPG) (Guideline): **"may perform manual therapy, directed at the cervical spine and upper extremity, for individuals with mild to moderate CTS in the short term"**
- ☐ **Orthosis + stretching** (Erickson 2019 JOSPT CTS CPG) (Guideline): **"may use a combined orthotic/stretching program in individuals with mild to moderate CTS who do not have thenar atrophy and have normal 2-point discrimination"**

B · THUMB CMC OA MOBILISATION & MANUAL OEDEMA MOBILISATION

CMC & OEDEMA

THUMB CMC OA

Magni 2022

- ☐ **Manual therapy + exercise (RCT, 60 pts aged 70–90)** (Villafañe 2013 JOSPT) [M]: **reduced pain vs sham; limited value for pressure pain threshold, pinch and grip**
- ☐ **Mobilisation (SR/MA, 8 RCTs)** (Magni 2022 Hand Ther) [L]: **pain SMD 0.53 (0.03 to 1), pinch SMD 0.35 (0.03 to 0.7) — statistically but not clinically significant**

HAND OEDEMA

Miller 2017

- ☐ **Manual oedema mobilisation (SR, 10 studies)** (Miller 2017 J Hand Ther) [L]: **low-to-moderate evidence supports manual oedema mobilisation with standard therapy for problematic hand oedema**
- ☐ **Technique** [E]: proximal clearing first, light skin-traction strokes toward the axilla, elevation and active pumping of the digits; re-measure with figure-of-eight / volumetry

C · DRY NEEDLING SAFETY & ADVERSE EVENTS

SAFETY SCREEN

- ☐ **Adverse event rates:** Brady 2014 [M]: mild AEs 19.18 %; upper risk rate for significant AEs ≤ 0.04 %; bruising 7.55 %, bleeding 4.65 % • Boyce 2020 [M]: minor AE 36.7 %; major AE < .1 % (prolonged symptom aggravation .03 %, fainting .02 %, forgotten needles .01 %)
- ☐ **Serious risks & screening:** Halle 2016 [E]: pneumothorax, cardiac tamponade, haematoma, infection, CNS injury; filament needles 32–36 gauge (.25–.20 mm) • Kearns 2019 [E]: anticoagulants / coagulopathy — caution — possible increased incidence or severity of haematoma and bruising

D · PRE-INTERVENTION SAFETY SCREEN, INFORMED CONSENT & MANUAL THERAPY LOG

CONSENT

Contraindications / Precautions Screened [E]: ☐ Fracture / scaphoid excluded or cleared ☐ No infection / Kanavel signs ☐ No acute CRPS flare with allodynia ☐ Post-op orders checked with surgeon

CONSENT OBTAINED FOR: ☐ Wrist / carpal / CMC mobilisation ☐ Cervical / upper-limb manual therapy ☐ Neural & tendon gliding ☐ Manual oedema mobilisation

Patient / Guardian Signature _____ **Clinician** _____ **Date** _____

DATE	TECHNIQUE (MOB / NEURAL / OEDEMA)	JOINT / DIRECTION / DOSAGE	IMMEDIATE NPRS / ROM	ADVERSE EVENTS

MANUAL THERAPY RESPONSE · GLIDE DIRECTION · HOME SELF-MOBILISATION INSTRUCTIONS

A · EXERCISE, RECOVERY, CRPS EXPOSURE, RETURN TO WORK & JOSPT CTS CPG OUTCOME MEASURES

EXERCISE & RTW

EXERCISE & RECOVERY

Østerås 2017

- ☐ **Hand OA exercise (Cochrane)** (Østerås 2017 Cochrane) [L]: **pain SMD −0.27 (−0.47 to −0.07), 5 trials 381 pts; function SMD −0.28 (−0.58 to 0.02); stiffness SMD −0.36 (−0.58 to −0.15); low quality**
- ☐ **Rheumatoid hand — SARAH (RCT, n = 490)** (Lamb 2015 Lancet) [H]: **tailored exercises: MHQ function +4·3 (1·5–7·1) at 12 months**
- ☐ **DRF recovery (cohort, n = 129)** (MacDermid 2003 BMC Musculoskelet Disord) [M]: **by 3 months median 'no pain at rest', worst pain 3/10; most recovery within six months; hardest tasks: carrying ten pounds, pushing up from a chair**
- ☐ **CRPS-I — exposure in vivo (RCT, n = 46)** (den Hollander 2016 Pain) [M]: **graded exposure more effective than pain-contingent care for disability in patients with pain-related fear**

RETURN TO WORK & OUTCOME MEASURES

JOSPT 2019

- ☐ **Hand / wrist trauma (cohort, n = 383)** (Neutel 2019 J Hand Surg Eur) [M]: **time off work linked to job type, diagnosis, complication, blaming someone else, gender**
- ☐ **Occupational upper-limb injury (cohort)** (Bai 2022 Front Med) [L]: **107/163 (65.6%) back at work at 1 year**
- ☐ **JOSPT CTS CPG — outcome measures** (Erickson 2019 JOSPT CTS CPG) (Guideline): **"should use the CTQ-SSS to assess symptoms and the … CTQ-FS or the … DASH questionnaire to assess function"; "should not use grip strength when assessing short-term (less than 3 months) change … following CTR surgery"; "should not use lateral pinch strength as an outcome measure"**

B · PAIN-MONITORING MODEL & ACTIVITY MODIFICATION

TRAFFIC LIGHT

- ☐ **Pain-monitoring model:** Sprague 2021 [E]: **pain should not exceed 5/10 on the NPRS during or immediately after activity and should return to pre-activity levels by the following morning** • Silbernagel 2007 [M]: continued tendon loading with a pain-monitoring model showed no negative effects
- ☐ **Origin of the zones:** Cavaggion 2024 [E]: low (0–2), acceptable (2–5) and high (5–10) risk zones were based on clinical experience

Low 0–2 (Safe) Acceptable 2–5 (Monitor) High 5–10 (Excessive)

Activity Guidance (E):

AVOID	MODIFY	OK

C · TENDON & NERVE GLIDING AND WRIST PROPRIOCEPTION PROGRESSION

PROGRESSION

- ☐ **Tendon gliding** [E]: straight hand → hook fist → full fist → tabletop → straight fist; smooth, pain-free excursion
- ☐ **Nerve gliding** [E]: median / ulnar sliders before tensioners; stop if paraesthesia persists after the set
- ☐ **Proprioception & loading** [E]: joint position sense → dart-thrower's motion → weight-bearing → perturbation → work / sport tasks

D · 12-WEEK WRIST & HAND PROGRESSIVE LOADING LOG

12-WEEK LOG

EXERCISE / DRILL	WK 1	WK 2	WK 3	WK 4	WK 5	WK 6	WK 7	WK 8	WK 9	WK 10	WK 11	WK 12
Tendon / Nerve Gliding												
Grip & Pinch Strength												
Wrist Ext / Flex / RD / UD Loading												
Pronation / Supination Load												
Proprioception & Weight-Bearing												
Work / Sport Task Simulation												

HOME EXERCISE PRESCRIPTION · WORK CONDITIONING · RETURN-TO-ACTIVITY NOTES

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

Longitudinal Session-by-Session Tracking Flowsheet

Patient _____ 09 / 09

#	DATE	NPRS	FLEX / EXT	RD / UD	GRIP / PINCH	EXERCISE & LOADING DELIVERED	ORTHOSIS / MANUAL / EPA	IRRITABILITY	PROM	PATIENT SIGNATURE	INIT
1						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
2						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
3						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
4						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
5						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
6						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
7						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
8						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
9						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
10						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
11						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			
12						Glide | Grip | Load | Prop | Task	Orth | Mob | Oedema | EPA	Low | Mod | High			

DISCHARGE MILESTONES (POPULATION-SPECIFIC MCID & MDC BENCHMARKS) & DISCHARGE SUMMARY

DISCHARGE

- ☐ PRWE [L]: **Surgical DRF** (Huyke-Hernández 2023): **MCID 43.1 (6 weeks), 56.0 (12 weeks)** · **DRF at 6 months** (Mahdavi 2022): **pain or function ≥ 18/50 → screen for catastrophizing, distress and fear**
- ☐ QuickDASH / DASH [M]: **Upper limb** (Franchignoni 2014): **QuickDASH MDC90 12.85, MCID 15.91; DASH MDC90 10.81, MCID 10.83**
- ☐ MHQ [L]: **Older adults after DRF** (Jayaram 2021): **interpret domain scores rather than the summary score**
- ☐ BCTQ [L]: **Women with CTS** (López-de-Uralde-Villanueva 2024): **MCID 0.64 symptom severity, 0.23 functional status**
- ☐ NPRS [L]: **Women with CTS** (López-de-Uralde-Villanueva 2024): **MCID 1.5 mean pain, 2.5 worst pain**
- ☐ Grip [L]: **same device every visit** (Mutalib 2022) · after CTR: **"should not use grip strength when assessing short-term (less than 3 months) change"** (Erickson 2019, Guideline)

Final PRWE / QuickDASH / BCTQ _____ **Final Grip / Pinch (kg)** _____ **Final Flex / Ext / RD / UD** _____ **Disposition** ☐ Self-Mgmt / RTW ☐ Hand Surgeon Refer

Session 6 checkpoint: Re-score the main PROM against the MCID / MDC already on the sheet. If there is no change, re-assess the diagnosis, the dose or referral (E).

DISCHARGE SUMMARY · LONG-TERM SELF-MANAGEMENT · RETURN-TO-WORK / SPORT CLEARANCE

Discharging Clinician _____ **Sig.** _____